

GP487 Web Mapping Report:
Climate Change and Water Scarcity in Lethbridge, Alberta

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Description of the Community, Issue, and User Group

The City of Lethbridge is a growing community in southern Alberta, with a population of 98,406 in 2021 and a projected population of 115,000 by 2026 (“Census profile”, 2023; “Lethbridge, Alberta”, n.d.). The city’s economy was traditionally based on agriculture, but has diversified into a major hub for southern Alberta’s industrial and commercial activities in recent years as well (“About Lethbridge”, n.d.). Lethbridge spans a total area of 127.19 km², divided by the Oldman River, and annually experiences an average maximum and minimum temperature of 12.8°C and -1.1°C, respectively (“About Lethbridge”, n.d.). Annual precipitation averages 380.2mm, with around 264 dry days per year, making Lethbridge the 11th driest city in Canada (“About Lethbridge”, n.d.).

Water scarcity is a large issue in Lethbridge. The city depends on the Oldman River and its reservoir system for municipal water supply (“Spring Water”, 2025). The three main reservoirs that feed the Oldman watershed are the Oldman Reservoir, the St. Mary Reservoir, and the Waterton Reservoir (“Spring Water”, 2025). In recent years, the region has experienced below-average mountain snowpack and precipitation, largely reducing runoff into the Oldman River system (“Spring Water”, 2025). Consequently, the Oldman Reservoir (the main reservoir feeding Lethbridge) was only approximately 65% full in 2025 (“Spring Water”, 2025). The entire Oldman River Basin has also been provincially designated as Drought Stage 3: Severe Drought in recent years, and Lethbridge has implemented voluntary water restrictions to reduce water waste (“Spring Water”, 2025). In the Prairie provinces, water scarcity is a major issue due to high agricultural irrigation demands (Dryden, 2021), and Southern Alberta is particularly dry, with Lethbridge receiving about 330 days of sunlight per year (“Location spotlight”, 2025). Across the province, 100% of available water sources have been allocated, creating vulnerability for both municipalities and industries (Dryden, 2021).

In the South Saskatchewan River Watershed, which includes the Oldman River Basin, rising temperatures are directly intensifying water scarcity. Warmer conditions can increase evapotranspiration rates and water demand from both natural ecosystems and agriculture (Mousavi et al., 2024a). These temperatures also cause snow events to occur later and melt during winter, reducing the quantity of snowmelt entering the reservoirs Lethbridge relies on (Mousavi et al., 2024a). Climate change is projected to greatly increase drought frequency and severity in the coming years, largely due to reduced precipitation and increased evaporation (Mousavi et al., 2024b). The locations and extent of the area most affected by drought will also change, shifting further east from the Rocky Mountains and covering ~76% more land by the end of the century (Mousavi et al., 2024b).

Overall, climate change is a primary cause of water scarcity in Lethbridge. This project will illustrate this relationship by presenting a time series comparison of key climate factors with indicators of water scarcity over the last decade, with a focus on the climate change factors of surface temperature, precipitation levels, and drought conditions. Temporally visualizing these trends in Lethbridge's reservoirs will help demonstrate how climate change impacts one of the most critical facets of the city's water distribution system.

The user group for this project is residents of the City of Lethbridge. We plan to increase public awareness of the connection between climate change and local water issues, as local impacts can be less visible, making it difficult for residents to understand challenges within their own community. For example, residents of Lethbridge experience impacts of water scarcity daily, particularly through recurring drought conditions. The mapping platform is therefore designed to be educational rather than analytical, targeting users with little or no experience using web map applications and directly connecting their lived experiences to quantitative and visual data. A study surveying local drought experiences found that individual perceptions have been largely overlooked, with limited understanding of how people perceive drought risk (Boyce et al., 2025). While residents can share their experiences through surveys to broaden day-to-day perspectives, providing accessible information can, in turn, complement future water scarcity research as well.

A web map is well-suited for this project topic and user group, since interactive web mapping makes information more accessible than static reports. Users can drive their own discovery of trends over different time periods, as well as focus on locations of particular interest, making conclusions more personally meaningful. By visualizing climate variables alongside water scarcity indicators within the Lethbridge area, residents can directly see how environmental change intersects with the places that are important to them. Users will be able to engage with this issue without needing to request datasets, attend council meetings, or rely on technical reports that might be difficult to understand. This open-access, localized approach will help residents better understand the reasoning behind voluntary water restrictions and support community engagement with city-wide water conservation plans.

Data Sources and Processing

Table 1: Data Sources

Category	Name and Link	Description
Background Info	City Boundary	City boundary of Lethbridge. Retrieved from the Lethbridge open data catalogue.

Background Info	Lakes and Rivers	Hydrographical features in Southern Alberta. Retrieved from the Alberta open data catalogue.
Climate Change (temp)	Maximum Temperature	Maximum surface temperature measurements in Canada. Retrieved from the Government of Canada.
Climate Change (precip)	Accumulated Precipitation	Precipitation measurements in Canada. Retrieved from the Government of Canada.
Climate Change (drought)	Canadian Drought Monitor	Drought classifications in Canada. Retrieved from the Government of Canada. • D0 (Abnormally Dry) – represents an event that occurs once every 3-5 years; • D1 (Moderate Drought) – represents an event that occurs every 5-10 years; • D2 (Severe Drought) – represents an event that occurs every 10-20 years; • D3 (Extreme Drought) – represents an event that occurs every 20-25 years; and • D4 (Exceptional Drought) – represents an event that occurs every 50 years.
Water Scarcity (level)	Oldman River Water Level	Daily water level measurements in the Oldman River. Retrieved from the Government of Canada.
Water Scarcity (level)	Oldman Reservoir Water Level	Daily water level measurements in the Oldman Reservoir. Retrieved from the Government of Canada.
Water Scarcity (level)	St. Mary Reservoir Water Level	Daily water level measurements in the St. Mary Reservoir. Retrieved from the Government of Canada.
Water Scarcity (level)	Waterton Reservoir Water Level	Daily water level measurements in the Waterton Reservoir. Retrieved from the Government of Canada.

The data sources have been listed in the table above. The following will discuss the processing and insertion of each data source in the Experience Builder and Leaflet apps.

City Boundary

This source is a polygon shapefile that illustrates the boundary of the City of Lethbridge. It was downloaded from the the City of Lethbridge's open data catalogue, then uploaded to ArcGIS Online (AGOL) for use in Experience Builder and placed in a data folder to be used in Leaflet.

Lakes and Rivers

This source displays hydrographical feature lines and polygons in Southern Alberta. It was downloaded as GeoJSON file type from the Alberta open data catalogue ESRI REST services. MapShaper (mapshaper.com) was then used to simplify the GeoJSON via weighted area visvalingam to speed up loading times. Finally, the file was uploaded to AGOL for use in Experience Builder and placed in a data folder to be used in Leaflet.

Maximum Temperature

The climate change factor of temperature is defined in this dataset as the maximum temperature (Celcius) over a given time period. This dataset was retrieved from the Government of Canada. The layer was saved to AGOL via the source's ArcGIS Online Map Viewer for the Experience Builder app. For the Leaflet app, it is called dynamically via the source's ESRI REST services JavaScript API.

Precipitation

The climate change factor of precipitation is defined in this dataset as accumulated precipitation measurements (mm) over a given time period. This dataset was retrieved from the Government of Canada. The layer was saved to AGOL via the source's ArcGIS Online Map Viewer for the Experience Builder app. For the Leaflet app, it is called dynamically via the source's ESRI REST services JavaScript API.

Drought

The climate change factor of drought is defined in this dataset as the Canadian Drought Monitor. This data provide various drought classification levels that were collected from the Government of Canada, which is defined in table 1. The layer was saved to AGOL via the source's ArcGIS Online Map Viewer for the Experience Builder app. For the Leaflet app, it is called dynamically via the source's ESRI REST services JavaScript API.

Water Level

Water level data was collected from all of the four focus locations: Oldman River, Oldman Reservoir, St. Mary Reservoir, and Waterton Reservoir. All datasets were retrieved from the Government of Canada. They were downloaded as CSV files containing the water levels for the particular location over each month and year. Rows containing years beyond the period of study

(2013-2024) were removed. Additionally, columns containing irrelevant factors such as water flow were removed. Lastly, a column containing the coordinates of the reservoir (obtained from the same source, but originally not in the CSV directly) was added. Finally, the cleaned CSVs were uploaded to AGOL for Experience Builder and placed in a data folder to be used in Leaflet.

Relative Water Level

Additional processing was done to calculate the relative water level percentage for each location, comparing each month's measurements to the entire given time period (2013-2024). Using a Python script, the historical lowest and highest values for each location over the given years were found. The Oldman River had a low value of 0.27 m and high value of 2.89 m; Oldman Reservoir had a low of 1094.28 m and high of 1118.43 m; St. Mary Reservoir had a low of 1079.08 m and high of 1103.47 m; and Waterton Reservoir had a low of 1167.30 m and high of 1185.09 m. Finally, the relative water level percent was calculated at each point in time using the following equation:

$$\text{relative percent} = (\text{current level} - \text{low level}) / (\text{high level} - \text{low level})$$

These values were then used to create concentric circle symbology for each key location, representing how high the water level was at the currently selected time.

Description of Applications

Both the ArcGIS Online (Experience Builder) and Leaflet (JavaScript) applications are interactive maps that provide the user with an environmental monitoring viewer examining several climate change-related factors that impact water scarcity in Lethbridge, Alberta, from 2013–2024. Both maps compare trends across the three primary map layers: Maximum Temperature (°C), Drought Monitor, and Accumulated Precipitation (mm) in relation to four water reservoirs: the Oldman River, Waterton Reservoir, St. Mary's Reservoir, and the Oldman Reservoir.

Launch Page

Link: <https://env-gis-srv1.uwaterloo.ca/w2026/gp487/group5/launch.html>

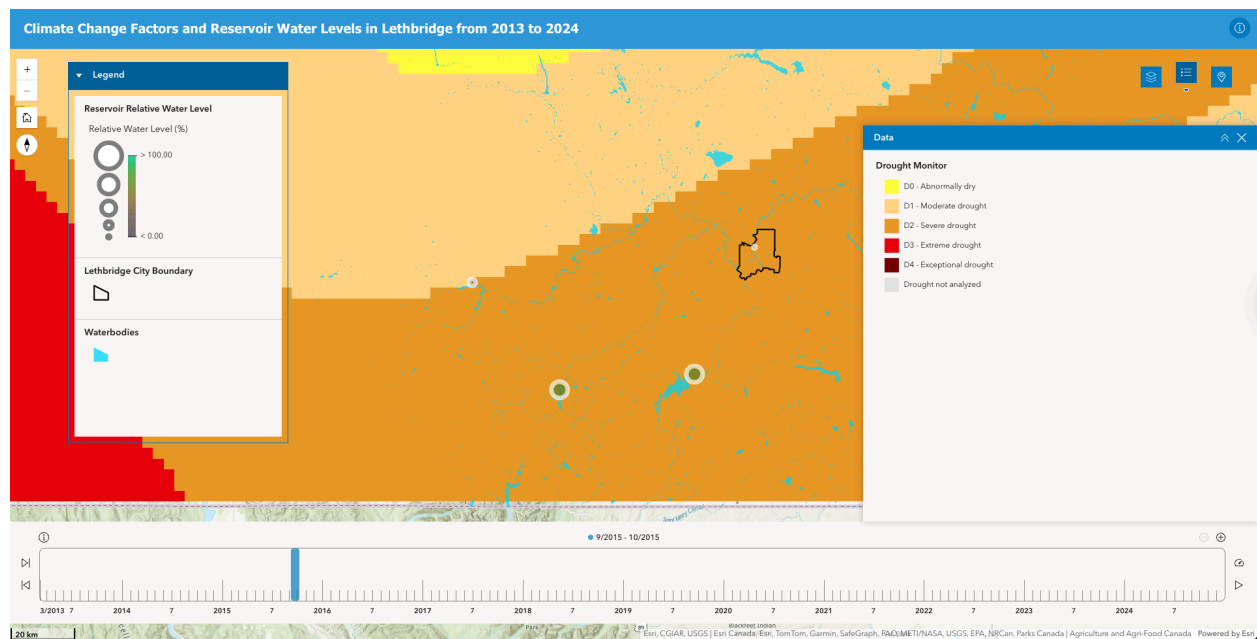
The launch page exists to give background context on Lethbridge and the issues of water scarcity and climate change. It also includes quick access links to each of the ArcGIS Online and Leaflet web applications.

ArcGIS Online

Link: <https://experience.arcgis.com/experience/0f5b26f078484696922f990fc9566d0d>

The map layout resembles a generally easy-to-navigate web map, including a north arrow, zoom controls, a scale, and a terrain basemap. A time slider from 2013–2024 appears at the bottom of the map, along with a drop-down legend showing the relative water percentage at each reservoir and the Lethbridge city boundary. Additional interactive features include turning on multiple climate-factor layers simultaneously, adjusting their transparency for comparison, and a select and zoom function for each reservoir location, which opens a pop-up with additional attribute information. Three tabs in the top right corner include ‘Map Layers’, ‘Data Layers’, and ‘View Reservoir’. The map layers provide raster datasets that can be toggled on or off, with their symbology displayed under the data layers panel for a better understanding of the map visual. Figure 1 is an example snapshot of the Experience Builder app displaying ‘D3 - Extreme drought’ conditions under the Drought Monitor map layer for August to September 2023, which was selected using the time slider at the bottom.

Figure 1: Experience Builder Map Interface Snapshot



Leaflet

Link: <https://env-gis-srv1.uwaterloo.ca/w2026/gp487/group5/leaflet/index.html>

The Leaflet interface is a user-friendly map layout that includes standard map elements such as a zoom function, north arrow, scale bar, and a time slider displaying data from 2013 to 2025. The legend pane shows symbology for lakes and rivers, the Lethbridge city boundary, reservoir-related water level (%), and one of our three selected climate change factors from the dropdown: precipitation, temperature, or the drought monitor. A unique feature of the map is that

clicking on a reservoir allows the user to view a graph of water levels over time, as shown in Figure 3. Figure 2 shows a snapshot of the Leaflet app displaying “D3 – Extreme drought” conditions under the Drought Monitor layer for August to September 2023, selected using the time slider at the bottom.

Figure 2: Leaflet Map Interface Snapshot

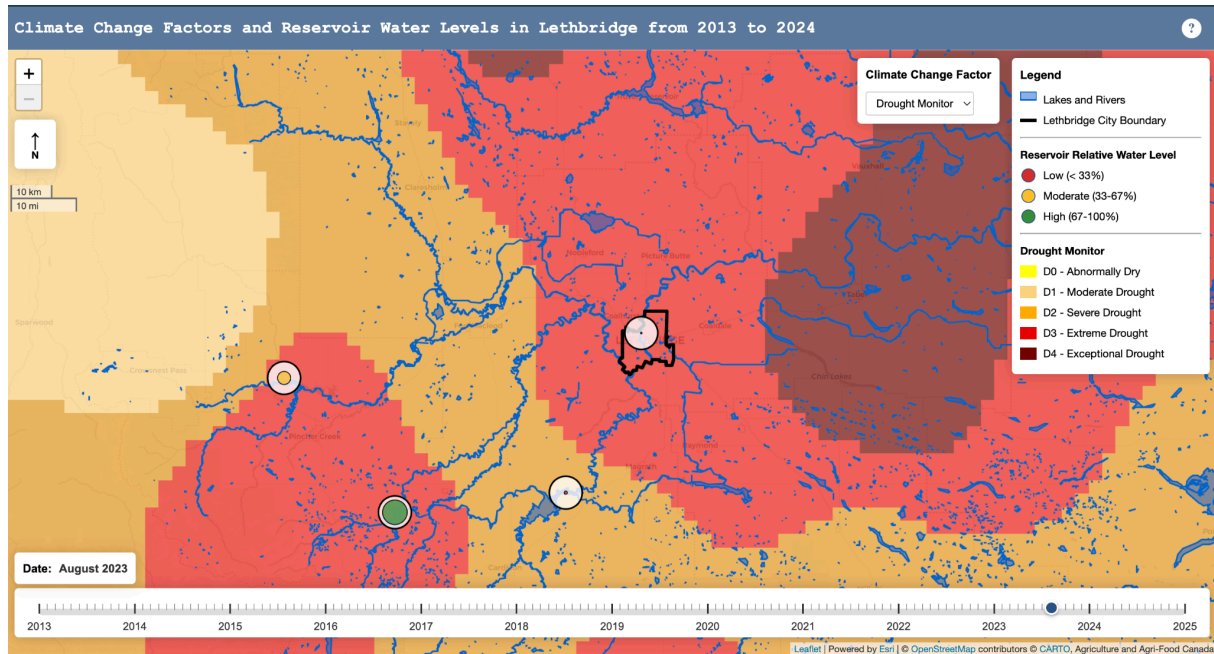
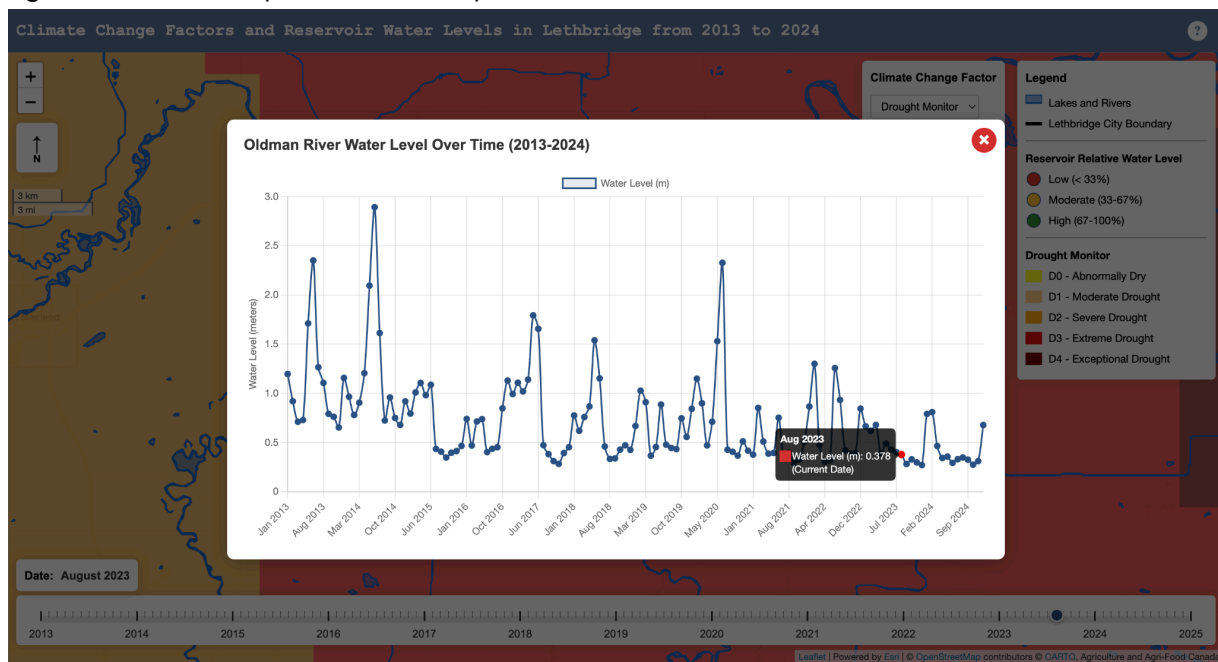


Figure 3: Leaflet Graph Function Snapshot



Evaluation of Web Mapping Options

This section will evaluate various web mapping options that are most appropriate in illustrating how climate change factors impact water scarcity in Lethbridge, considering the needs of both the municipal government and end-users. It will focus on two mapping options that are ideal for the topic: ArcGIS Online (Experience Builder) and Leaflet (JavaScript). Each option will be evaluated by both qualitative and quantitative metrics that relate to how the app is able to fit organizational requirements, development environments, functionality, and performance.

1. Organizational Requirements

ArcGIS Online:

This option is commonly used in Canadian municipal environments. Many municipal governments utilize Esri products for zoning, environment, transportation, and storing data for analysis and public demonstration. The City of Lethbridge is in the same situation, since it has obtained Esri licensing and thus an ArcGIS Online solution can be easily integrated into existing municipal GIS workflows. Personnel would likely already be familiar with using Esri software, speeding up the learning process. Esri documentation and tutorials is also very extensive, providing clear and direct professional support. Furthermore, AGOL is a web mapping software that uses a browser-based interface. Thus, it doesn't require programming expertise to create and maintain the application and datasets.

Leaflet:

This option is a leading open-source JavaScript library for creating mobile-friendly interactive maps ("An open-source", n.d.). It doesn't require any licensing or subscription, which makes it financially attractive. However, its implementation requires more advanced technical expertise, including coding in JavaScript and HTML, web development, and potential server-side configuration. While there is extensive support for programming in many online sources, current city personnel may not have any experience in this field which would make the learning curve steeper. Continuous maintenance would also be required from knowledgeable developers, causing higher costs in the long run despite being free up front.

2. Development Environment

ArcGIS Online:

It is simple to perform common GIS tasks in AGOL, such as publishing climate change data, symbolizing different water levels, adding pop-ups for river features, and adding time visualizations. Experience Builder allows the user to immediately insert pre-built tools into the map. It also includes documentation and tutorials online for users to follow, which are very

well-structured as Esri's user community is large and active. Moreover, advanced functions can be configured without coding by utilizing Dashboards and Experience Builder. This creates a friendly development environment for non-developers in the department to create maps with associated data.

Leaflet:

Leaflet's development environment presumes users to have a solid programming background, despite documentation and examples being easily accessible. Basic tutorials for implementing simple features exist, but professional knowledge in writing code would be required for creating more complex custom components. This complexity increases development time and lowers accessibility for municipal staff that have no or limited experience in programming.

3. Functionality

ArcGIS Online:

AGOL provides all the capabilities needed for the City of Lethbridge to perform analysis on the given climate change and water scarcity factors. For instance, it allows time-enabled layers for illustrating data changes over time, layer filtering and querying for various climate change factors, and support for common spatial data formats such as shapefiles and GeoJSON. The platform supports integration of spatial and temporal data in a unified interface, enabling users to explore relationships between climate variables and water indicators effectively.

Leaflet:

Leaflet can support practically any functionality, as it is used directly through JavaScript. Coding is very versatile, since you can custom-design any feature you'd like and import additional libraries, such as for charting or map display. There is also good API support for GIS analysis through Esri. Therefore, it can achieve the same level of mapping and beyond when compared to AGOL, but requires more effort to develop and is limited by technical capability.

4. Performance

ArcGIS Online:

The loading and refresh speeds in Experience Builder are reasonably fast (around 10 seconds to open and 1 second for functionality) because the platform is created for efficient public access. Any significant limitations on high volumes of data is unlikely, as this project includes large rasters that were handled appropriately.

Leaflet:

The performance of coding-based web applications completely relies on the implementation quality and hosting infrastructure. In this project, the loading and refresh speed is fast (1 second to open and functionality). Any significant limitations on high volumes of data is unlikely, as this project includes large rasters that were handled appropriately.

Overall Conclusion

From a municipal governance and public perspective, ArcGIS Online offers the most balanced and practical solution. While Leaflet provides greater flexibility and higher performance, its reliance on advanced development expertise and ongoing technical support reduces its suitability for long-term municipal deployment. AGOL aligns most closely with the City of Lethbridge's operational capacity, public engagement goals, and need for transparent, accessible communication regarding climate-related water challenges. Additionally, AGOL's flexibility and performance are still suitable for this particular project, reducing the benefits of Leaflet. Thus, this evaluation supports the selection of AGOL as the preferred web mapping platform for this initiative.

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